

Year 9 Science – Booklet 1: Physics

Work & Energy — ANSWERS

Quick Test (p.3)

1. Any five of: heat (thermal) energy, electrical energy, light energy, chemical energy, gravitational potential energy, nuclear energy, sound energy, kinetic energy, elastic potential energy.
2. Chemical energy, elastic potential energy, gravitational potential energy.
3. Gravitational potential energy.
4. Kinetic energy.
5. Electrical energy is transferred into heat (thermal) energy and kinetic energy (of the moving air), with a little sound energy too.
6. Sound energy is transferred into electrical energy.

Section A – Multiple Choice (5 marks)

1. d) sound
2. c) chemical
3. b) elastic potential energy
4. c) electrical energy
5. a) kinetic energy

Section B (14 marks)

1. Matching (9 marks – first one given)

- “The energy an object has because it is moving.” → **Kinetic energy**
- “The energy produced by reactions in the centre of an atom.” → **Nuclear energy**
- “Vibrating objects are sources of this energy.” → **Sound energy**
- “Forms of energy that are waiting to be used.” → **Stored energy**
- “Food is an example of this.” → **Chemical energy**
- “This energy is available every time a current flows.” → **Electrical energy**
- “Winding up a spring will give it this type of energy.” → **Elastic potential energy**
- “Most of our energy on the Earth begins as this.” → **Light energy**
- “When one type of energy changes into another type of energy.” → **Energy transfer**

2. Word order (5 marks)

- a). Burning a candle changes chemical energy into heat and light energy.
- b). A loudspeaker changes electrical energy into sound energy.
- c). A battery stores chemical energy.
- d). Water which is flowing has kinetic energy.
- e). Chemical energy, elastic potential energy and gravitational potential energy are all forms of stored energy.

Section C (18 marks)

1. Energy changes in the pictures (a car, a CD player & loudspeakers, a bulb & battery)

- a). The car is changing **chemical** energy into **kinetic** energy and **heat** energy.
- b). The CD player and loudspeakers are changing **electrical** energy into **sound** energy and **heat** energy.
- c). The light bulb is changing **electrical** energy into **light** energy and **heat** energy.

2. Apparatus and energy sources

- a) i). A microphone.
- a) ii). A catapult (or a stretched elastic band/spring).
- a) iii). A battery (or a fuel cell).
- b). From food (chemical energy stored in food).
- c). From sunlight (light energy from the Sun, used in photosynthesis).

3. Explain the energy changes

- a). Heat (thermal) energy from the Sun is transferred to the water, giving it enough energy to evaporate (turn from liquid to water vapour) and rise into the air, where it cools and condenses to form clouds.
- b). As the rainwater flows down the mountainside, its store of gravitational potential energy decreases and is transferred into kinetic energy as it speeds up, flowing back down into the lake.